Supplier For. Various Exp. Buyer ✓ Seller ✗

COST SHEET

1. Cost Sheet

① Accrual Basis

It is a statement which shows the break-up and build-up of costs for a particular period (functional classification).

2. Statement of Cost/Cost Sheet

Particulars	Total Cost (₹)	Cost per unit (₹)
Opening stock of Raw Material	✓	
Add: Purchases	✓	
Less: Closing stock of Raw Material	(✓)	
Add: Carriage/Freight inward	✓	
Less: Raw material purchase return	(✓)	
Less: Sale value of scrap of raw material (Question Specified)	(✓)	
Direct Material Consumed	✓	
Add: Direct labour cost incurred ➢ Wages ✓ ➢ Bonus ✓ ➢ Allowances ✓ ➢ Overtime ✓ ➢ Employer contribution to PF/ESI/SSS ✓ ➢ Any other benefit ✓	✓	
Add: Direct expenses or chargeable expenses incurred ➢ Royalty for production ✓ ➢ Cost of utilities such as power & fuel, steam etc. ✓ ➢ Fee for technical know-how ✓ ➢ Cost of product/service specific design or drawing ✓ ➢ Cost of product/service specific software ✓ ➢ Amortized cost of moulds, patterns, patents etc. ✓ ➢ Job charges paid to job workers ✓ ➢ Hire charges paid for hiring specific equipment ✓ ➢ Other expenses which are directly related with production ✓	✓	
Prime Cost	✓	
Add: Factory/Work Overheads	✓	
Gross Factory/Work Cost	✓	FG + WIP
Add: Opening stock of WIP	✓	
Less: Closing stock of WIP	(✓)	
Net Factory/Work cost	✓	FG Produced

Particulars	Total Cost (₹)	Cost per unit (₹)
Add: Quality control cost	✓	
Add: Research and Development cost	✓	
Add: Administrative overheads (related to production)	✓	
Add: Packing cost (primary)	✓	
Less: Credit for recoveries/Scrap/Defectives/By-Product	(✓)	
Cost of Production	✓	FG Produced
Add: Opening stock of finished goods	✓	
Less: Closing stock of finished goods	(✓)	
Cost of Goods Sold	✓	FG Sold
Add: Administrative overheads (general)	✓	
Add: Selling and distribution overheads	✓	
Add: Interest on borrowed funds	✓	
Cost of Sales	✓	FG Sold
Add: Profit	✓	
Sales	✓	FG Sold

3. Points to Remember (PTR)

(A) Loss or Gain

Loss or Gain
- Normal — Charge to consumer — By reducing quantity and increasing price per unit
- Abnormal — Charge to owner — By transferring to P&L

(B) Certain expenses not appear in cost sheet:

- Goodwill or preliminary expenses written off
- Income tax
- Loss on sale of assets or investment
- Cost pertaining to or arising out of a pandemic e.g. COVID-19
- Penalty, fines, damages etc.

(C) Work = Factory

Work Overheads = Factory Overheads

Work Cost = Net Factory Cost

Op. St. + Current Prod. FG

(D) Cost of goods available for sale = Opening stock of FG + Cost of Production (COP)

4. Treatment of Expenses

Expenses	Treatment
Insurance on stock of Raw material & WIP	Factory OHs
Stores Related Expenses Stores Consumed = Opening + Purchases – Closing	Factory OHs
Warehouse or Godown Expenses	Selling & Distribution OHs
Loose tools written off	Factory OHs
Bank charges	Administration OHs
Salesmen commission	Selling & Distribution OHs
Cost of Samples	Selling & Distribution OHs
Audit Fee	Administration OHs
General Expenses	Administration OHs
Counting House Salaries	Administration OHs
Production planning expenses in office	Administration OHs related to Prod.
Director's fees	Administration OHs
Fee for exhibition participation	Selling & Distribution OHs
Pollution control expenses	Factory OHs
Carriage on raw material return	Factory OHs
Bad Debts	Ignore
Packaging	
(A) Primary Packaging	Add after NFC
(B) Secondary Packaging	Selling & Distribution OHs
GST	
(A) GST Output	Add after sales value

(B) GST Input	
(C) ITC Available	Ignore
(D) ITC Not Available	Add to the cost of item to which it relates
Custom Duty	Add to the cost of item to which it relates
Discount	
(A) Trade Discount	Deduct if not already deducted
(B) Cash Discount	Ignore
(C) Other discount or discount on sales	Selling & Distribution OHs
Waste/Scrap	
(A) Scrap for which amount is received on sale	
(1) Related to raw material	Deduct from raw material consumed
(2) Arises during production $\boxed{\text{Default}}$	Deduct after NFC
(B) Scrap for which disposal cost is to be incurred	Add to factory OHs
Defectives	
(a) Sold as it is at discount	Deduct after NFC
(b) Goods are rectified by incurring rectification cost	Add rectification cost to Factory OHs

5. Administration Overheads

If administration overheads is ___% of NFC or If administration overheads is ₹ ___ per unit produce then in both situation consider them as related to production.

RM → $\boxed{\text{Fac.}}$ → FG

6. Conversion Cost

It is the cost to convert raw material into finished goods. It is sum total of direct labour, direct expenses and factory overheads.

→ $\boxed{\text{DL + D.Exp. + Fac. OH}}$

Question – 1

SK ltd. has the following expenditures for the year ended 31st March:

Particulars	Amount (₹)	Amount (₹)
Raw materials purchased → RMC		10,00,00,000
GST paid on the above purchases @18% (eligible for input tax credit) → Ignore		1,80,00,000
Freight inwards → RMC		11,20,600
Wages paid to factory workers → DW		29,20,000
Contribution made towards employees' PF and ESI → DW		3,60,000
Production bonus paid to factory workers → DW		2,90,000
Royalty paid for production → D. Exp.		1,72,600
Amount paid for power & fuel → D. Exp.		4,62,000
Amount paid for purchase of moulds and patterns (life is equivalent to two years production) → D. Exp. $= \frac{8.96L}{2} = 4.48L$		8,96,000
Job charges paid to job workers → D. Exp.		8,12,000
Stores and spares consumed → Fac. OH		1,12,000
Depreciation on:		
Factory building → Fac. OH	84,000	
Office building → Adm. OH	56,000	
Plant & Machinery → Fac. OH	1,26,000	
Delivery vehicles → S&D OH	86,000	3,52,000
Salary paid to supervisors → Fac. OH		1,26,000
Repairs & maintenance paid for:		
Plant & Machinery → Fac. OH	48,000	
Sales office building → S&D OH	18,000	
Vehicles used by directors → A. OH	19,600	85,600
Insurance premium paid for:		
Plant & Machinery → Fac. OH	31,200	
Factory building → Fac. OH	18,100	
Stock of raw materials & WIP → Fac. OH	36,000	85,300
Expenses paid for quality control check activities → QCC		19,600
Salary paid to quality control staffs → QCC		96,200
Research & development cost paid for improvement in production process → R&D		18.200
Expenses paid for pollution control and engineering & maintenance → Fac. OH		26,600
Expenses paid for administration of factory work → A. OH (P)		1,18,600
Salary paid to functional managers:		
Production control → A. OH (P)	9,60,000	

Finance & accounts → A. OH	9,18,000	
Sales & Marketing → S&D OH	10,12,000	28,90,000
Salary paid to General Manager → A.OH		12,56,000
Packaging cost paid for:		
Primary packing necessary to maintain quality → Primary Pack.	96,000	
For re-distribution of finished goods → S&D OH	1,12,000	2,08,000
Wages of employees engaged in distribution of goods → S&D OH		7,20,000
Fee paid to auditors → A.OH		1,80,000
Fee paid to legal advisors → A.OH		1,20,000
Fee paid to independent directors → A.OH		2,20,000
Performance bonus paid to sales staff → S&D OH		1,80,000
Value of stock as on 1st April of last year		
Raw materials → op. RM	18,00,000	
Work-in-process → op. WIP	9,20,000	
Finished goods → op. FG	11,00,000	38,20,000
Value of stock as on 31st March of current year		
Raw materials → Cl. RM	9,60,000	
Work-in-process → Cl. WIP	8,70,000	
Finished goods → Cl. FG	18,00,000	36,30,000 Deduct After NFC

Amount realized by selling of scrap and waste generated during manufacturing process is ₹ 86,000.

From the above data you are required to prepare statement of cost for the year ended 31st March, showing (i) prime cost, (ii) factory cost, (iii) cost of production, (iv) cost of goods sold and (v) cost of sales.

Solution

Particulars		Amount (₹)
Opening stock of raw material		18,00,000
Add: Raw material purchases		10,00,00,000
Less: Closing stock of raw material		(9,60,000)
Add: Freight inwards		11,20,600
Raw material consumed		10,19,60,600
Direct Labour:		
Wages paid to factory workers	29,20,000	
Contribution to PF & ESI	3,60,000	
Production bonus paid to factory workers	2,90,000	35,70,000
Direct Expenses:		
Royalty paid for production	1,72,600	
Amount paid for power & fuel	4,62,500	
Amortised cost of moulds and patterns	4,48,000	

Job charges paid to job workers	8,12,000	35,70,000
Prime Cost		10,74,25,200
Factory overheads:		
Stores and spares consumed	1,12,000	
Depreciation on factory building	84,000	
Depreciation on plant & machinery	1,26,000	
Repairs & maintenance for plant & machinery	48,000	
Insurance premium paid for plant & machinery	31,200	
Insurance premium paid for factory building	18,100	
Insurance premium paid for stock of raw material	36,000	
Salary paid to supervisors	1,26,000	
Expenses paid for pollution control	26,600	6,07,900
Gross Factory cost		10,83,33,100
Add: Opening WIP		9,20,000
Less: Closing WIP		(8,70,000)
Net Factory cost		10,80,83,100
Quality control cost:		
Expenses paid for quality control check	19,600	
Salary paid to quality control staff	96,200	1,15,800
Research and development cost paid		18,200
Administrative overheads related to production		
Expenses paid for administration	1,18,600	
Salary paid to production control manager	9,60,000	10,78,600
Less: Realisable value on sale of scrap		(86,000)
Add: Primary packaging cost		96,000
Cost of production		10,93,05,700
Add: Opening stock of finished goods		11,00,000
Less: Closing stock of finished goods		(18,000,000)
Cost of goods sold		10,86,05,700
Administrative overheads:		
Depreciation on office building	56,000	
Repairs & maintenance paid for vehicles for directors	19,600	
Salary paid to manager-finance and accounts	9,18,000	
Salary paid to general manager	12,56,000	
Fee paid to auditors	1,80,000	
Fee paid to legal advisors	1,20,000	
Fee paid to independent directors	2,20,000	27,69,600
Selling and distribution overheads		

Repairs & maintenance paid to sales office building	18,000	
Salary paid to manager – sales & marketing	10,12,,000	
Performance bonus paid to sales staffs	1,80,000	
Depreciation on delivery vehicles	86,000	
Packaging cost paid for re-distribution	1,12,000	
Wages of employees engaged in distribution of goods	7,20,000	21,28,000
Cost of Sales		11,35,03,300

Question – 2

The following data relates to manufacturing of a standard product during the month of the March:

Particulars	Amount (in ₹)
Stock of Raw material as on 01-03 → Op. RM	80,000
Work in progress as on 01-03 → Op. WIP	50,000
Purchase of raw material → RMC	2,00,000
Carriage inwards → RMC	20,000
Direct wages → DW	1,20,000
Cost of special drawing → D. Exp.	30,000
Hire charges paid for Plant → D. Exp.	24,000
Return of Raw Material → RMC	40,000
Carriage on return → Fac. OH	6,000
Expenses for participation in Industrial exhibition → S&D OH	8,000
Legal charges → A. OH	2,500
Salary to office staff → A. OH	25,000
Maintenance of office building → A. OH	2,000
Depreciation on Delivery Van → S&D OH	6,000
Warehousing charges → S&D OH	1,500
Stock of Raw material as on 31-03 → Cl. RM	30,000
Stock of Work in Progress as on 31-03 → Cl. WIP	24,000

RMC = 80000 + 200000 + 20000 − 40000 − 30000 = 2,30,000

PC = 230000 + 120000 + 30000 + 24000 = 404000

F.OH
- Store overheads on material are 10% of material consumed. → 10% × 2,30,000 = 23000

F.OHs
- Factory overheads are 20% of the prime cost → 20% × 4,04,000 = 80800
- 10% of the output was rejected and a sum of ₹ 5,000 was realized on sale of scrap. → After NFC
- 10% of the finished product was found to be defective and the defective products were rectified at an additional expenditure which is equivalent to 20% of proportionate direct wages.
- The total output was 8,000 units during the month. ↳ $20\% \times (\underbrace{\underbrace{120000 \times 90\%}_{\text{Good units DW}} \times 10\%}_{\text{Def. unit wages}}) = 2160$ Fac. OH

You are required to prepare a cost sheet for the above period showing the:

(i) Cost of raw material consumed

(ii) Prime cost

(iii)Work cost
(iv)Cost of production
(v) Cost of sales

Solution **Cost Sheet**

Particulars			**Amount (₹)**
Opening stock of raw material			80,000
Add: Raw material purchases			2,00,000
Add: Carriage inward			20,000
Less: Return of raw material			(40,000)
Less: Closing stock of raw material			(30,000)
		Raw Material consumed	2,30,000
Direct wages			1,20,000
Direct Expenses:	Cost of special drawing	30,000	
	Hire charges paid for plant	24,000	54,000
		Prime Cost	4,04,000
Stores Overheads (10% × 2,30,000)			23,000
Carriage on return of raw material			6,000
Factory overheads (20% × 4,04,000)			80,800
Rectification cost of defectives (1,20,000 × 90% × 10% × 20%)			2,160
		Gross Factory Cost	5,15,960
Add: Opening WIP			50,000
Less: Closing WIP			(24,000)
		Net Factory Cost	5,41,960
Less: Scrap sale			(5,000)
		Cost of Production/COGS	5,36,960
Administration Overheads:			
	Legal charges	2,500	
	Salary to office staff	25,000	
	Maintenance of office building	2,000	29,500
Selling & Distribution Overheads:			
	Expenses for participation in industrial exhibition	8,000	
	Warehousing charges	1,500	
	Depreciation on Delivery Van	6,000	15,500
		Cost of Sales	5,81,960

Question – 3	

A fire occurred in the factory premises on October 31. The accounting records have been destroyed.

COP = Cost Av. For Sale − Op. FG
= 555775 − 37750 = 518025

Certain accounting records were kept in another building. They reveal the following for the period September 1 to October 31:

(a) Direct materials purchased → RMC — ₹ 2,50,000
(b) Work in progress inventory (1 Sep) → Op. WIP — ₹ 40,000
(c) Direct material inventory (1 Sep) → Op. RM — ₹ 20,000
(d) Finished goods inventory (1 Sep) → Op. FG — ₹ 37,750
(e) Indirect manufacturing costs → F. OH — 40% of conversion cost
(f) Sales revenue → Sales — ₹ 7,50,000
(g) Direct Manufacturing labour → DL — ₹ 2,22,250
(h) Prime costs → PC — ₹ 3,97,750
(i) Gross margin percentage based on revenues → Pft — 30%
(j) Cost of goods available for sale → $\boxed{\text{Op. FG + COP}}$ = ₹ 5,55,775

Sales	750000
(−) Pft @ 30%	
COS/COGS	525000
(+) Cl. FG (B/F)	30775
(−) Op. FG	(37750)
COP	518025

COP/NFC	518025
(+) Cl. WIP (B/F)	67892
(−) Op. WIP	(40000)
GFC	545917

(397750 + 148167)

The cost is fully covered by insurance. The insurance company wants to know the historical cost of the inventories as the basis for negotiating a settlement, although the settlement is actually to be based on replacement cost, not historical cost.

Let For. OH = y
y = 40% x (222250 + 0 + y)
y = 88900 + (0.4)(y)
y = 88900/0.6 = 148167

PC	397750
(−) DL	222250
DMC	175500
(+) Cl. St. (B/F)	94500
(−) Op. St.	(20000)
RM Purch.	250000

Required:

(a) Finished goods inventory 31 October
(b) Work in process inventory 31 October
(c) Direct material inventory 31 October

Solution

Statement of cost and sales

Particulars	Amount
Opening stock of material	20,000
Add: Purchases	2,50,000
Less: Closing stock of material(bal. fig.)	(94,500)
Direct material consumed (bal. fig.)	1,75,500
Add: Direct Labour	2,22,250
Add: Direct Expenses	-
Prime Cost (given)	3,97,750
Add: Factory Overheads (working note-1)	1,48,167
Gross Factory Cost	5,45,917
Add: Opening WIP	40,000
Goods Processed during the period	5,85,917
Less: Closing WIP(bal. fig.)	(67,892)
Net Factory Cost/COP (bal. fig.)	5,18,025
Add: Opening stock of FG	37,750
Cost of goods available for sale (given)	5,55,775
Less: Closing stock of FG(bal. fig.)	(30,775)
Cost of goods sold	5,25,000
Add: Administration Overheads	-
Add: Selling & Distribution Overheads	-

Particulars	Amount
Cost of Sales (Bal. fig.)	5,25,000
Add: Profit (7,50,000 × 30%)	2,25,000
Sales	7,50,000

Working note - 1

Let factory overheads = x

Thus, x = 40% (2,22,250 + 0 + x)

x = 1,48,167

7. Valuation of Stock

Stock can be valued either on FIFO basis or LIFO basis or Weighted average method. Unless otherwise provided FIFO method will be used for valuation of stock.

According to FIFO Method,

Cost P.U.

Value of cl. stock of raw material = $\boxed{\frac{\textit{Amount of raw material purchased}}{\textit{Raw material purchase quantity}}} \times \textit{Cl. stock raw material units}$

Value of closing stock of finished goods = $\boxed{\frac{\textit{Cost of Production}}{\textit{Units produced}}} \times \textit{Closing stock finished goods units}$

Cost P.U.

According to Weighted Average Method,

Cost P.U.

Value of cl. stock of raw material = $\boxed{\frac{\textit{RM purchase} + \textit{Op. Stock of RM}}{\textit{RM purchase units} + \textit{Op. stock RM units}}} \times \textit{Cl. stock raw mat. units}$

Value of closing stock of finished goods = $\boxed{\frac{\textit{Cost of Production} + \textit{Op. stock FG}}{\textit{Units produced} + \textit{Op. stock FG units}}} \times \textit{Closing stock FG units}$

Cost P.U.

For raw material,

⇒ Raw material consumed units = Op. stock RM units + RM Purchase units – Cl. stock RM

For finished goods,

⇒ Finished goods units sold = Op. stock FG units + FG units produced – Cl. stock FG units

If there is NIL Opening stock

Cost per unit of finished goods = $\frac{\textit{Cost of Production}}{\textit{Units produced}} = \frac{\textit{Cost of Goods Sold}}{\textit{Units Sold}}$

8. Calculation of per unit data and vice versa

(A) For calculating per unit data

- Divide all values by Units Produced upto cost of production
- Divide all values by Units Sold from COGS and onwards

(B) If factory overheads or administration overheads (production related) per unit is given then multiply it with number of units produced to get total value.

(C) If administration overheads (general) and selling and distribution overheads per unit is given then multiply it with number of units sold to get total value.

Question – 4

From the following particulars, you are required to prepare monthly cost sheet of SK Ltd.:

	Amount (₹)
Opening Inventories	
- Raw materials → op. RM	12,00,000
- Work-in-process → op. WIP	18,00,000
- Finished goods (10,000 units) → op. FG	9,60,000
Closing inventories	
- Raw materials → Cl. RM	14,00,000
- Work-in-process → Cl. WIP	16,04,000
- Finished goods → Cl. FG	? → $\frac{\text{CoP}}{194000} \times 44000$
Raw material purchased → RMC	1,44,00,000
GST paid on raw materials purchased (ITC available) → Ignore	7,20,000
Wages paid to production workers → DL	36,64,000
Expenses paid for utilities → D. Exp.	1,45,600
Office and administration expenses paid → A. OH	26,52,000
Travelling allowance paid to office staffs → A. OH	1,21,000
Selling expenses → S&D OH	6,46,000

Machine hours worked – 21,600 hours
Machine hour rate - ₹ 8.00 per hour
→ f. OH OH = 21600 × 8 = 172800

Units sold – 1,60,000
Units produced 1,94,000
Desired profit – 15% on sales

Sales = op. + Prod. − Cl.
160000 = 10000 + 194000 − Cl.
Cl. St. FG = 44000 units

Solution

Cost sheet of SK Ltd. for month of …..
Units produced – 1,94,000
Units sold – 1,60,000

Particulars	(₹)	Cost per unit (₹)
Raw materials purchased	1,44,00,000	
Add: Opening value of raw materials	12,00,000	
Less: Closing value of raw materials	(14,00,000)	
Materials consumed	1,42,00,000	73.19
Wages paid to production workers	36,64,000	18.89
Expenses paid for utilities	1,45,600	0.75
Prime Cost	**1,80,09,600**	**92.83**
Factory overheads (₹ 8 × 2,600 hours)	1,72,800	
Add: Opening value of W-I-P	18,00,000	
Less: Closing value of W-I-P	(16,04,000)	
Cost of Production	**1,83,78,400** → $\frac{18378400}{194000}$ =	**94.73**
Add: Value of opening finished stock	9,60,000	
	(41,68,120)	

Less: Value of closing finished stock (₹ 94.73 × 44,000)		
Cost of Goods Sold	**1,51,70,280**	**94.81**
Office and administration expenses paid	26,52,000	16.58
Travelling allowance paid to office staffs	1,21,000	0.75
Selling expenses	6,46,000	4.04
Cost of Sales	1,85,89,280	116.18
Add: Profit	32,80,461	20.50
Sales	**2,18,69,741**	**136.68**

Question – 5

XYZ a manufacturing firm, has revealed following information for September, 2019:

	1st September	30th September
	₹	₹
Raw Materials	2,42,000	2,92,000
Works-in-progress	2,00,000	5,00,000

The firm incurred following expenses for a targeted production of 1,00,000 units during the month:

units Sold = 0 + 100000 − 5000 − (4%×1L)
= 91000

	₹
Consumable Stores and spares of factory – F. OH	3,50,000
Research and development cost for process improvements R&D	2,50,000
Quality control cost QCC	2,00,000
Packing cost (secondary) per unit of goods sold S&D	2
Lease rent of production asset → F. OH	2,00,000
Administrative Expenses (General) → A. OH	2,24,000
Selling and distribution Expenses → S&D	4,13,000
Finished goods (opening)	Nil
Finished goods (closing) → Cl. FG	5,000 units

COGS		7826000
(+) Cl. FG	(78.26L / 91000 × 5000)	430000
(−) Op. FG		–
COP		82,56,000
(−) QCC		(200000)
(−) R&D		(250000)
(+) Def. (4000×61)		2,44,000
NFC		8050000
(+) Cl. WIP		500000
(−) Op. WIP		(200000)
GFC		8350000
(−) F.OH		(550000)
PC		7800000
(−) DW		(2600000)
DMC	(78L ÷ 150%)	5200000
(+) Cl. RM		292000
(−) Op. RM		(242000)
		52.50L

Defective output which is 4% of targeted production, realizes ₹ 61 per unit. Closing stock is valued at cost of production (excluding administrative expenses). Cost of goods sold, excluding administrative expenses amounts to ₹ 78,26,000. Direct employees' cost is ½ of cost of material consumed. Selling price of the output is ₹ 110 per unit.

You are required to:

(i) Calculate the value of material purchased

(ii) Prepare cost sheet showing the profit earned by the firm

COGS		7826000
(+) A. OH		224000
(+) S&D		413000
(+) Sec. Pac.	(2×91000)	182000
COS		
(+) Pft.		
Sales	(91000×110)	

Solution

Statement of cost and profit

Particulars	**Amount**
Opening stock of material	2,42,000
Add: Purchases (w.n. - 2)	52,50,000
Less: Closing stock of material	(2,92,000)
Direct material consumed	52,00,000
Add: Direct Labour (w.n. - 2)	26,00,000
Add: Direct Expenses	-
Prime Cost (bal. fig.)	78,00,000
Add: Factory Overheads	
Consumable stores and spares of factory	3,50,000
Lease rent of production asset	2,00,000
Gross Factory Cost (bal. fig.)	83,50,000
Add: Opening WIP	2,00,000
Less: Closing WIP	(5,00,000)
Net Factory Cost (bal. fig.)	80,50,000
Add: Quality Control Cost	2,00,000
Add: Research and development Cost	2,50,000
Less: Sale of defective goods (1,00,000 × 4% × 61)	(2,44,000)
Cost of Production (bal. fig.)	82,56,000
Add: Opening stock of FG	-
Less: Closing stock of FG (w.n. – 1)	(4,30,000)
Cost of goods sold	78,26,000
Add: Administration overheads	2,24,000
Add: Packaging cost (Secondary) (2 × 91,000)	1,82,000
Add: Selling & Distribution overheads	4,13,000
Cost of Sales	86,45,000
Add: Profit (bal. fig.)	13,65,000
[(1,00,000 – 4,000 – 5,000) × 110] **Sales**	1,00,10,000

Working Notes:

(1) Since there is no opening stock, so the entire cost of goods sold is out of cost of production only.

Thus, cost per unit of goods produced = $\frac{78{,}26{,}000}{1{,}00{,}000-5{,}000-4{,}000}$ = ₹ 86

Therefore, closing stock of finished goods = 86 × 5,000 = ₹ 4,30,000

(2) Let raw material purchase = y

Thus, raw material consumed = 2,42,000 + y – 2,92,000 = y – 50,000

Direct wages = ½ × (y – 50,000) = 0.5y – 25,000

Prime cost = y – 50,000 + 0.5y – 25,000

78,00,000 = 1.5y – 75,000

1.5y = 78,75,000

y = 52,50,000

Raw material purchased = y = ₹ 52,50,000

9. Recovery Rate

It is the rate which is used to recover/absorb/charge overheads from the products being manufactured or services being provided.

Unless otherwise provided, following basis will be used for recovery of overheads:

Factory overheads	→	Direct Labour
Administration overheads	→	NFC or work cost
Selling and distribution overheads	→	NFC or work cost

Question – 6

The following figures are available from the books of SK Co. for the year 31st March:

	₹		₹
Materials:		Profit for the year → Pft.	12,180
Stock on 1st April → op. RM	2,000	Selling overhead → S&D	10,500
Stock on 31st March → CL. RM	4,000	Factory overhead → F.OH	9,000
Purchases → RMC	20,000	Administration overhead → A.OH	8,400
Wages → DW	15,000		

(a) Prepare a cost sheet showing prime cost, work cost, cost of production, cost of sales and sales.

(b) In April, the factory receives an order for a job which will require materials ₹ 2,400 and wages ₹ 1,500. Ascertain the sale price of the job if the factory intends to earn a profit 10% higher than the percentage of profit earned in year ending on 31st March. Assume that the factory overhead has gone up by 16(2/3)% and selling overhead has gone down by 20% after 31st March. Further assume that factory overhead is recovered as a percentage of the wages and administration and selling overhead as a percentage of works cost.

DM	2400
DW	1500
PC	3900
(+) F.OH (70% × 1500)	1050
GFC/NFC/COP/WC	4950
(+) A.OH (4950 × 20%)	990
(+) S&D (4950 × 20%)	990
COS	6930
(+) Pft. (6930 × 22%)	1525
SP	8455

Solution

Statement of Cost and Profit

Particulars	Amount (₹)
Opening stock of material	2,000
Add: Purchases	20,000
Less: Closing stock of material	(4,000)
Direct material consumed	18,000

Add: Direct wages	15,000
Prime cost	33,000
Add: Factory overhead	9,000
GFC/NFC/COP/COGS	42,000
Add: Administration overhead	8,400
Add: Selling overhead	10,500
Cost of Sales	60,900
Add: Profit	12,180
Sales	73,080

Req. P/f % = $\frac{(12180+10\%)}{60900} \times 100 = 22\%$ of COS

F.OH Rate = $\frac{(9000+16\frac{2}{3}\%)}{15000} \times 100 = 70\%$ of DW

A.OH Rate = $\frac{8400}{42000} \times 100 = 20\%$ of NFC

S&D OH Rate = $\frac{(10500-20\%)}{42000} \times 100 = 20\%$ of NFC

Calculation of Recovery Rates

Factory overheads as % of direct wages $= \frac{(9{,}000+16.6666666\%)}{15{,}000} \times 100 = 70\%$ of direct wags

Administration overheads as % of NFC $= \frac{8{,}400}{42{,}000} \times 100 = 20\%$ of NFC

Selling overheads as % of NFC $= \frac{(10{,}500-20\%)}{42{,}000} \times 100 = 20\%$ of NFC

Profit as % of Cost of sales $= \frac{(12{,}180+10\%)}{60{,}900} \times 100 = 22\%$ of Cost of sales

Statement of calculation of selling price of Job

Particulars		**Amount (₹)**
Direct Material		2,400
Direct wages		1,500
Prime Cost		3,900
Add: Factory overheads	(70% × 1,500)	1,050
GFC/NFC/COP/COGS		4,950
Add: Administration overheads	(20% × 4,950)	990
Add: Selling overheads	(20% × 4,950)	990
Cost of sales		6,930
Add: Profit	(22% × 4,950)	1,525
Sales		8,455

10. Change in Cost Effects

Total Cost = No. Of units × Cost per unit

	Total VC	Total FC
Quantity Effect	Yes	No
Price Effect	Yes	Yes

Relation between: → All directly related except wages & Efficiency

(A) Quantity and Variable cost → : Direct relation

(B) Price and Variable cost → : Direct relation

(C) Price and Fixed cost → : Direct relation
(D) Wages and Efficiency → : Indirect relation

Direct Relation
↳ Adj. in Nr.
Inverse Relation
↳ Adj. in Dr.

Unless otherwise provided, following points are to be assumed:

(a) VC per unit will remain same
(b) Total FC will remain same
(c) All direct cost are considered to be variable in nature
(d) All overheads are considered to be fixed in nature

Let Ex. units = 100
(+) 60% = 60
160
(-) Eff. Dec. @10% = 16
New units = 144

Question – 7	

A factory incurred the following expenditure during the year:

		₹
Direct material consumed		12,00,000 × 144/100 × 94/100 = 1624320
Manufacturing wages		7,00,000 × 144/100 × 100/90 = 1120000
Manufacturing overheads:		
Fixed	3,60,000 × 120/100 = 432000	
Variable	2,50,000 × 144/100 × 160/100 = 576000	6,10,000
		25,10,000

COS = 3752320
(+) Pft. =
SP = 375320 / 90%
= 4169244

In the next year, following changes are expected in production and cost of production.

(a) Production will increase due to recruitment of 60% more workers in the factory.
(b) Overall efficiency will decline by 10% on account of recruitment of new workers.
(c) There will be an increase of 20% in fixed overhead and 60% in variable overhead.
(d) The cost of direct material will be decreased by 6%.
(e) The company desire to earn a profit of 10% on selling price.

Ascertain the cost of production and selling price.

Solution

Let existing production units	100
Add: Increase due to recruitment of worker(100 × 60%)	60
	160
Less: Decline due to efficiency (160 × 10%)	16
New Production units	144

Statement of cost and sale

Particulars	Working	Amount (₹)
Direct material	$\left(12{,}00{,}000 \times \frac{144}{100} \times \frac{94}{100}\right)$	16,24,320

Direct wages	$\left(7,00,000 \times \frac{144}{100} \times \frac{100}{90}\right)$	11,20,000
Prime Cost		**27,44,320**
(+) Fixed manufacturing overheads	$\left(3,60,000 \times \frac{120}{100}\right)$	4,32,000
(+) Variable manufacturing overheads	$\left(2,50,000 \times \frac{144}{100} \times \frac{160}{100}\right)$	5,76,000
Cost of Sales		**37,52320**
(+) Profit	(Bal. fig.)	4,16,924
Sales	(37,52,320 ÷ 90%)	**41,69,244**

Question – 8

A factory's normal capacity is 1,20,000 units per annum. The estimated costs of production are as under:

(a) Direct material ₹ 3 per unit; direct labour ₹ 2 per unit (Subject to a minimum of ₹ 12,000 p.m.)

(b) Indirect expenses—Fixed ₹ 1,60,000 per annum: Variable ₹ 2 per unit; Semi-variable ₹ 60,000 upto 50% capacity and additional ₹ 20,000 for every 20% increase in capacity.

(c) Each unit of raw material yields scrap which is sold at the rate of 20 paise per unit.

– 50%
+ 20000 – 20%
+ 20000 – 20%
100000 – 90%

The factory worked at 50% capacity for the first three months but it was expected that it would work @ 80% capacity for the remaining 9 months. During the first three months, the selling price per unit was ₹ 12. What should be the price in the remaining nine months to produce a total profit of ₹ 2,18,000?

Solution

Statement of Cost

Particulars	First 3 months	Bal. 9 months
Level of operation	50%	80%
Units	$1,20,000 \times \frac{50}{100} \times \frac{3}{12} =$ 15,000	$1,20,000 \times \frac{80}{100} \times \frac{9}{12} =$ 72,000
Direct material @ ₹ 3 p.u.	45,000	2,16,000
Direct wages	$\left\{\begin{matrix} 15,000 \times 2 \\ or \\ 12,000 \times 3 \end{matrix}\right\}$ 36,000	$\left\{\begin{matrix} 72,000 \times 2 \\ or \\ 12,000 \times 9 \end{matrix}\right\}$ 1,44,000
Fixed expenses	$1,60,000 \times \frac{3}{12} =$ 40,000	$1,60,000 \times \frac{9}{12} =$ 1,20,000
Variable expenses @ ₹ 2 p.u.	30,000	1,44,000
Semi-variable expenses	$60,000 \times \frac{3}{12} =$ 15,000	$(60,000 + 20,000 + 20,000) \times \frac{9}{12}$ = 75,000
(-) Scrap @ ₹ 0.20 p.u.	(3,000)	(14,400)
Total Cost	**1,63,000**	**6,84,600**

P/t. = (15000×12) − 163000
= 17000

(-) RP (218000−17000) = 201000

Sale 885600
units 72000
SP 12.30

Statement of Calculation of Selling Price for Remaining 9 Months

Sales for first 3 months (15,000 × 12)	1,80,000
Less: Cost for first 3 months	1,63,000
Profit for first 3 months	17,000
Annual Target profit	2,18,000
Profit require from remaining 9 months	2,01,000
Add: Cost for remaining 9 months	6,84,600
Sales for remaining 9 months	8,85,600
Units for remaining 9 months	72,000
Selling price for remaining 9 months	12.30

Question – 9

PNME Ltd. manufactures two types of masks – 'disposal Masks' and 'Cloth Masks'. The cost data for the year ended 31st March, 2022 is as follows:

	₹
Direct materials	12,50,000
Direct wages	7,00,000
Production Overhead	4,00,000
Total	23,50,000

(50000 × 2) : (150000 × 1)
(50000 × 1) : (150000 × 0.6)
(50000 × 1) : (150000 × 1)

Mat. 500000
Lab. 250000
P. OH 100000
A. OH (1ℓ × 50%) 50000
COP 900000
(-) Cl. St. (9ℓ/50000 × 5000) (90000)
COGS 810000
(+) S&D (2 × 45000) 90000
COS 900000
(+) Pft. 6.75ℓ
Sale (35 × 45000) 15.75ℓ

It is further ascertained that:

- Direct material cost per unit of cloth Mask was twice as much of direct material cost per unit od disposal Mask
- Direct wages per unit for Disposal Mask were 60% of those for Cloth Mask
- Production overhead per unit was at same rate for both the types of the masks
- Administration overhead was 50% of Production overhead for each type of mask
- Selling cost was ₹ 2 per cloth mask
- Selling price was ₹ 35 per unit of cloth mask
- No. of units of cloth masks sold – 45,000 → Cl. St. = 5000
- No. of units of Production of
    - Cloth Masks : 50,000
    - Disposal Masks : 1,50,000

You are required to prepare a cost sheet for cloth masks showing:

(i) Cost per unit and total cost

(ii) Profit per unit and total profit

Solution

Preparation of Cost Sheet for Cloth Masks

No. of units produced = 50,000 units

No. of units sold = 45,000 units

Particulars	Per unit (₹)	Total (₹)
Direct materials (Working note (ii))	10.00	5,00,000
Direct wages (Working note (ii))	5.00	2,50,000
Prime cost	15.00	7,50,000
Production overhead (Working note (iii))	2.00	1,00,000
Factory Cost	17.00	8,50,000
Administration Overhead* (50% of Production Overhead)	1.00	50,000
Cost of production	18.00	9,00,000
Less: Closing stock (50,000 units – 45,000 units)	–	(90,000)
Cost of goods solid i.e. 45,000 units	18.00	8,10,00
Selling cost	2.00	90,000
Cost of sales/Total cost	20.00	9,00,000
Profit	15.00	6,75,000
Sales value (₹ 35 × 45,000 units)	35.00	15,75,000

Working Notes:

(i) Direct material cost per unit of Disposable Mask = M

Direct material cost per unit of Cloth Mask = 2M

Total direct material cost = (2M × 50,000) + (M × 1,50,000)

12,50,000 = M × 2,50,000

M = 12,50,000 ÷ 2,50,000 = ₹ 5

Thus, direct material cost per unit of cloth mask = 2 × 5 = ₹ 10

(ii) Direct wages per unit of Cloth Mask = W

Direct wages per unit Disposable Mask = 0.6W

So, (W × 50,000) + (0.6W × 1,50,000) = ₹ 7,00,000

W = ₹ 5 per unit

Therefore, Direct material Cost per unit of Cloth Mask = ₹ 5

(iii) Production overhead per unit = $\frac{4,00,000}{(50,000+1,50,000)}$ = ₹ 5

Production overhead for Cloth Mask = ₹ 2 × 50,000 units = ₹ 1,00,000

*Administration overhead is related to production overhead in the question and hence to be considered in cost of production only.

Question – 10	

MNP Limited have the capacity to produce 84,000 units of a product very month. Its prime cost per unit at various levels of production is as follows:

Level	Prime Cost per unit (₹)
10%	50
20%	48
30%	46
40%	$\boxed{44}$ Jan.
50%	42
60%	40

70%	38
80%	36
90%	34
100%	$\boxed{32}$ → Feb

Its prime cost consists of raw material consumed, direct wages and direct expenses in the ratio of 3:2:1. In the month of January 2024, the company worked at 40% capacity and raw material purchased amounting to ₹8,40,000. In the month of February 2024, the company worked at 100% capacity and raw material purchased for ₹16,46,400.

It is the policy of the company to maintain opening stock of raw material equal to 1/3 of closing stock of raw material. Factory overheads are recovered at 60% of direct wages cost. Fixed administration expenses (as part of production cost) and fixed selling and distribution expenses are ₹2,01,600 and ₹1,68,000 per month respectively. During the month of January 2024 company sold 33,600 units @ ₹68.8 per unit. The variable distribution cost amounts to ₹1.5 per unit sold.

The management of the company chalks out a plan for the month of February 2024 to sell its whole output @ ₹61 per unit by incurring following further expenditure:

→ 26250×4 = 105000

(i) Company sponsors a television programme on every Sunday at a cost of ₹26,250 per week. There are 4 Sundays in February 2024.

(ii) Hi-tea programme every month for its potential customers at a cost of ₹1,05,000. ✓

(iii) Special gift item costing ₹105 on sale of a dozen units. → $\frac{105}{12} \times (84000 \times 100\%) = 735000$

(iv) Lucky draws scheme is introduced every month by giving the first prize of ₹1,00,000; second prize of ₹80,000; third prize of ₹40,000 and four consolation prizes of ₹8,000 each.

12l + 0.80l + 0.40l + (4×0.08l) = 2.52l

Note: (In the month of February 2024, there is a significant saving in material cost per unit due to entry of new suppliers in the market and saving in per unit cost of Direct wages and Direct expenses due to introduction of new· policy by the management.)

Prepare a cost sheet for the month of January 2024 and February 2024 showing prime cost (with different elements of prime cost), factory cost, cost of production, total cost and profit earned.

Solution

op + 8.40l − Cl. = 739200

$(\frac{1}{3})(Cl.) + 8.40l - Cl. = 739200$

$(-\frac{2}{3})(Cl.) = -100800$

$Cl. = \frac{-100800}{-2/3} = 151200$

Cost Sheet

Particulars	January 2024	February 2024
84000×40% =	**33,600 Units**	**84,000 Units** ✓
Opening Stock of Raw Material (1/3)	50,400	1,51,200
Add: Purchases (Given)	8,40,000	16,46,400
Less: Closing stock of Raw Material	(1,51,200)	(4,53,600)
Direct materials consumed: 1478400×3/6 →	**7,39,200**	**13,44,000**
Direct Wages 1478400×2/6 →	4,92,800	8,96,000
Direct expenses 1478400×1/6 →	2,46,400	4,48,000
Prime Cost 33600×44 =	**14,78,400**	**26,88,000** = 84000×32
Factory overheads (60% of direct wages) →	2,95,680	→ 5,37,600

Factory / Works Cost	**17,74,080**	**32,25,600**
Add: Administration overhead (Production)	2,01,600	2,01,600
Cost of Production / Cost of goods sold	**19,75,680**	**34,27,200**
Add: Fixed selling and distribution Overhead	1,68,000	1,68,000
Variable distribution overheads (₹1.5 per unit)	50,400	1,26,000
Sponsorship cost	–	1,05,000
Hi tea programme	–	1,05,000
Special gifts (84,000 × 1/12 × 105)	–	7,35,000
Lucky draw prize*	–	2,52,000
Cost of sales / Total Cost	**21,94,080**	**49,18,200**
Profit (Balancing figure)	1,17,600	2,05,800
Sales revenue	**23,11,680**	**51,24,000**

***Lucky draw prize:**

	Amount (₹)
1st Prize	1,00,000
2nd Prize	80,000
3rd Prize	40,000
Consolation Prizes (4 × ₹8,000)	32,000
Total	**2,52,000**

Working note:

Calculation of opening and costing stock of Raw Material

January

Units Manufactured = 84,000 × 40% = 33,600 units
Prime Cost = 33,600 × 44 = ₹14,78,400
Raw Material consumed = ₹14,78,400 × 3/6 = ₹7,39,200
Raw Material purchase (given) = ₹8,40,000
Let closing stock of Raw Material be x
Opening stock of Raw Material be 1/3x
Opening Stock + Purchase – closing stock = Raw Material consumed
1/3x + ₹8,40,000 – x = ₹7,39,200
1/3x – x = ₹7,39,200 – ₹8,40,000
2/3x = ₹1,00,800
x = ₹1,51,200 (closing stock)
Opening stock = ₹1,51,200 × 1/3 = ₹50,400

February

Prime Cost = 84,000 × 32 = ₹26,88,000
Raw Material consumed = ₹26,88,000 × 3/6 = ₹13,44,000
Raw Material purchased (given) = ₹16,46,400
Opening Stock + Purchase – closing stock = Raw Material consumed
₹1,51,200 + ₹16,46,400 – closing stock = ₹13,44,000
Closing stock = ₹4,53,600